

```
In [1]: !pip install fastf1 matplotlib pandas
```

```
Collecting fastf1
  Downloading fastf1-3.8.3-py3-none-any.whl.metadata (5.1 kB)
Requirement already satisfied: matplotlib in
c:\users\user\anaconda3\lib\site-packages (3.10.6)
Requirement already satisfied: pandas in
c:\users\user\anaconda3\lib\site-packages (2.3.3)
Requirement already satisfied: cryptography in
c:\users\user\anaconda3\lib\site-packages (from fastf1) (46.0.3)
Requirement already satisfied: numpy<3.0.0,>=1.26.0 in
c:\users\user\anaconda3\lib\site-packages (from fastf1) (2.3.5)
Requirement already satisfied: platformdirs in
c:\users\user\anaconda3\lib\site-packages (from fastf1) (4.5.0)
Requirement already satisfied: pydantic in
c:\users\user\anaconda3\lib\site-packages (from fastf1) (2.12.4)
Requirement already satisfied: pyjwt in
c:\users\user\anaconda3\lib\site-packages (from fastf1) (2.10.1)
Requirement already satisfied: python-dateutil in
c:\users\user\anaconda3\lib\site-packages (from fastf1)
(2.9.0.post0)
Collecting rapidfuzz (from fastf1)
  Downloading rapidfuzz-3.14.5-cp313-cp313-win_amd64.whl.metadata
(12 kB)
Collecting requests-cache>=1.0.0 (from fastf1)
  Downloading requests_cache-1.3.3-py3-none-any.whl.metadata (9.4
kB)
Requirement already satisfied: requests>=2.30.0 in
c:\users\user\anaconda3\lib\site-packages (from fastf1) (2.32.5)
Requirement already satisfied: scipy<2.0.0,>=1.11.0 in
c:\users\user\anaconda3\lib\site-packages (from fastf1) (1.16.3)
Collecting signalrcore (from fastf1)
  Downloading signalrcore-1.0.2-py3-none-any.whl.metadata (13 kB)
Collecting timple>=0.1.6 (from fastf1)
  Downloading timple-0.1.8-py3-none-any.whl.metadata (2.0 kB)
Collecting websockets>=10.3 (from fastf1)
  Downloading websockets-16.0-cp313-cp313-win_amd64.whl.metadata
(7.0 kB)
Requirement already satisfied: contourpy>=1.0.1 in
c:\users\user\anaconda3\lib\site-packages (from matplotlib)
(1.3.3)
Requirement already satisfied: cycler>=0.10 in
c:\users\user\anaconda3\lib\site-packages (from matplotlib)
(0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in
c:\users\user\anaconda3\lib\site-packages (from matplotlib)
(4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in
c:\users\user\anaconda3\lib\site-packages (from matplotlib)
(1.4.9)
Requirement already satisfied: packaging>=20.0 in
c:\users\user\anaconda3\lib\site-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in
```

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c:\users\user\anaconda3\lib\site-packages (from matplotlib)
(12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in
c:\users\user\anaconda3\lib\site-packages (from matplotlib)
(3.2.5)
Requirement already satisfied: pytz>=2020.1 in
c:\users\user\anaconda3\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in
c:\users\user\anaconda3\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: six>=1.5 in
c:\users\user\anaconda3\lib\site-packages (from python-dateutil-
>fastf1) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
c:\users\user\anaconda3\lib\site-packages (from requests>=2.30.0-
>fastf1) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in
c:\users\user\anaconda3\lib\site-packages (from requests>=2.30.0-
>fastf1) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in
c:\users\user\anaconda3\lib\site-packages (from requests>=2.30.0-
>fastf1) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
c:\users\user\anaconda3\lib\site-packages (from requests>=2.30.0-
>fastf1) (2025.11.12)
Requirement already satisfied: attrs>=21.2 in
c:\users\user\anaconda3\lib\site-packages (from requests-
cache>=1.0.0->fastf1) (25.4.0)
Requirement already satisfied: cattrs>=22.2 in
c:\users\user\anaconda3\lib\site-packages (from requests-
cache>=1.0.0->fastf1) (25.3.0)
Collecting url-normalize>=2.0 (from requests-cache>=1.0.0->fastf1)
  Downloading url_normalize-3.0.0-py3-none-any.whl.metadata (7.5
kB)
Requirement already satisfied: typing-extensions>=4.14.0 in
c:\users\user\anaconda3\lib\site-packages (from cattrs>=22.2-
>requests-cache>=1.0.0->fastf1) (4.15.0)
Requirement already satisfied: cffi>=2.0.0 in
c:\users\user\anaconda3\lib\site-packages (from cryptography-
>fastf1) (2.0.0)
Requirement already satisfied: pycparser in
c:\users\user\anaconda3\lib\site-packages (from cffi>=2.0.0-
>cryptography->fastf1) (2.23)
Requirement already satisfied: annotated-types>=0.6.0 in
c:\users\user\anaconda3\lib\site-packages (from pydantic->fastf1)
(0.6.0)
Requirement already satisfied: pydantic-core==2.41.5 in
c:\users\user\anaconda3\lib\site-packages (from pydantic->fastf1)
(2.41.5)
Requirement already satisfied: typing-inspection>=0.4.2 in
c:\users\user\anaconda3\lib\site-packages (from pydantic->fastf1)
(0.4.2)
Collecting msgpack==1.1.2 (from signalrcore->fastf1)
  Downloading msgpack-1.1.2-cp313-cp313-win_amd64.whl.metadata
(8.4 kB)
Downloading fastf1-3.8.3-py3-none-any.whl (135 kB)
Downloading requests_cache-1.3.3-py3-none-any.whl (70 kB)
Downloading timple-0.1.8-py3-none-any.whl (17 kB)
```

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Downloading url_normalize-3.0.0-py3-none-any.whl (16 kB)
Downloading websockets-16.0-cp313-cp313-win_amd64.whl (178 kB)
Downloading rapidfuzz-3.14.5-cp313-cp313-win_amd64.whl (1.5 MB)
----- 0.0/1.5 MB ? eta --
-:--
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eta 0:00:02
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eta 0:00:02
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eta 0:00:02
----- 1.0/1.5 MB 616.9 kB/s
eta 0:00:01
----- 1.3/1.5 MB 662.9 kB/s
eta 0:00:01
----- 1.5/1.5 MB 690.0 kB/s
0:00:02
Downloading signalrcore-1.0.2-py3-none-any.whl (55 kB)
Downloading msgpack-1.1.2-cp313-cp313-win_amd64.whl (72 kB)
Installing collected packages: websockets, url-normalize,
rapidfuzz, msgpack, signalrcore, requests-cache, timple, fastf1

```

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----- 0/8 [websockets]
----- 0/8 [websockets]
----- 0/8 [websockets]
----- 1/8 [url-normalize]
----- 2/8 [rapidfuzz]
----- 2/8 [rapidfuzz]
Attempting uninstall: msgpack
----- 2/8 [rapidfuzz]
----- 3/8 [msgpack]
Found existing installation: msgpack 1.1.1
----- 3/8 [msgpack]
Uninstalling msgpack-1.1.1:
----- 3/8 [msgpack]
Successfully uninstalled msgpack-1.1.1
----- 3/8 [msgpack]
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----- 3/8 [msgpack]
----- 3/8 [msgpack]
----- 3/8 [msgpack]
----- 4/8 [signalrcore]
----- 5/8 [requests-cache]
----- 5/8 [requests-cache]
----- 7/8 [fastf1]
----- 7/8 [fastf1]
----- 8/8 [fastf1]
```

Successfully installed fastf1-3.8.3 msgpack-1.1.2 rapidfuzz-3.14.5
requests-cache-1.3.3 signalrcore-1.0.2 timple-0.1.8 url-normalize-
3.0.0 websockets-16.0

WARNING: Failed to remove contents in a temporary directory
'C:\Users\User\anaconda3\Lib\site-packages\~sgpack'.
You can safely remove it manually.

```
In [4]: import fastf1
from fastf1 import plotting
import matplotlib.pyplot as plt

# Προετοιμασία της βιβλιοθήκης για όμορφα γραφήματα F1
plotting.setup_mpl()

# 1. Φορτώνουμε το Grand Prix της Μεγάλης Βρετανίας 2024 (Qualifying)
# Σημείωση: Την πρώτη φορά θα πάρει λίγη ώρα γιατί κατεβάζει τα δεδομένα
session = fastf1.get_session(2024, 'Silverstone', 'Q')
session.load()

# 2. Παίρνουμε τον ταχύτερο γύρο του Hamilton (HAM)
ham_lap = session.laps.pick_driver('HAM').pick_fastest()

# 3. Παίρνουμε τα δεδομένα τηλεμετρίας για αυτόν τον γύρο
ham_telemetry = ham_lap.get_car_data().add_distance()

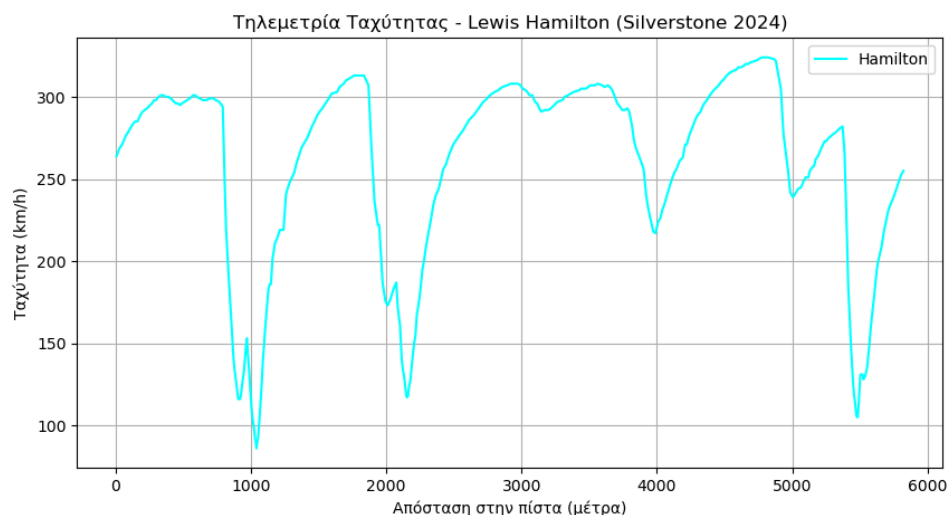
# 4. Σχεδιάζουμε το γράφημα: Απόσταση (X) και Ταχύτητα (Y)
plt.figure(figsize=(10, 5))
plt.plot(ham_telemetry['Distance'], ham_telemetry['Speed'],
label='Hamilton', color='cyan')

# Ονομασίες στους άξονες
plt.xlabel('Απόσταση στην πίστα (μέτρα)')
plt.ylabel('Ταχύτητα (km/h)')
plt.title('Τηλεμετρία Ταχύτητας - Lewis Hamilton (Silverstone 2024)')
plt.legend()
plt.grid(True)

# Εμφάνιση του γραφήματος
plt.show()
```

```
req          WARNING    DEFAULT CACHE ENABLED! (24.0 KB)
C:\Users\User\AppData\Local\Temp\fastf1
core          INFO      Loading data for British Grand Prix -
Qualifying [v3.8.3]
req          INFO      No cached data found for session_info.
Loading data...
_api         INFO      Fetching session info data...
req          INFO      Data has been written to cache!
req          INFO      No cached data found for driver_info.
Loading data...
_api         INFO      Fetching driver list...
req          INFO      Data has been written to cache!
req          INFO      No cached data found for
session_status_data. Loading data...
_api         INFO      Fetching session status data...
req          INFO      Data has been written to cache!
req          INFO      No cached data found for
track_status_data. Loading data...
_api         INFO      Fetching track status data...
req          INFO      Data has been written to cache!
req          INFO      No cached data found for
extended_timing_data. Loading data...
_api         INFO      Fetching timing data...
```

```
_api      INFO      Parsing timing data...
req       INFO      Data has been written to cache!
req       INFO      No cached data found for
timing_app_data. Loading data...
_api      INFO      Fetching timing app data...
req       INFO      Data has been written to cache!
core      INFO      Processing timing data...
core      WARNING    Fixed incorrect tyre stint information
for driver '18'
req       INFO      No cached data found for car_data.
Loading data...
_api      INFO      Fetching car data...
_api      INFO      Parsing car data...
req       INFO      Data has been written to cache!
req       INFO      No cached data found for position_data.
Loading data...
_api      INFO      Fetching position data...
_api      INFO      Parsing position data...
req       INFO      Data has been written to cache!
req       INFO      No cached data found for weather_data.
Loading data...
_api      INFO      Fetching weather data...
req       INFO      Data has been written to cache!
req       INFO      No cached data found for
race_control_messages. Loading data...
_api      INFO      Fetching race control messages...
req       INFO      Data has been written to cache!
core      INFO      Finished loading data for 20 drivers:
['63', '44', '4', '1', '81', '27', '55', '18', '23', '14',
'16', '2', '22', '24', '3', '77', '20', '31', '11', '10']
C:\Users\User\anaconda3\Lib\site-packages\fastf1\core.py:3175:
FutureWarning: pick_driver is deprecated and will be removed in
a future release. Use pick_drivers instead.
  warnings.warn(("pick_driver is deprecated and will be
removed"
```



```

In [2]: import fastf1
        from fastf1 import plotting
        import matplotlib.pyplot as plt

        # Προετοιμασία της βιβλιοθήκης για όμορφα γραφήματα F1
        plotting.setup_mpl()

        # Ενεργοποίηση της προσωρινής μνήμης (cache) για να κατεβαίνουν τα
        # δεδομένα γρήγορα
        fastf1.Cache.enable_cache('cache_dir')

        # 1. Φορτώνουμε το Grand Prix της Μεγάλης Βρετανίας 2024 (Qualifying)
        session = fastf1.get_session(2024, 'Silverstone', 'Q')
        session.load()

        # 2. Παίρνουμε τον ταχύτερο γύρο του Hamilton (HAM)
        ham_lap = session.laps.pick_driver('HAM').pick_fastest()

        # 3. Παίρνουμε τα δεδομένα τηλεμετρίας για αυτόν τον γύρο
        ham_telemetry = ham_lap.get_car_data().add_distance()

        # 4. Σχεδιάζουμε το γράφημα: Απόσταση (X) και Ταχύτητα (Y)
        plt.figure(figsize=(10, 5))
        plt.plot(ham_telemetry['Distance'], ham_telemetry['Speed'],
                 label='Hamilton', color='cyan')

        # Ονομασίες στους άξονες
        plt.xlabel('Απόσταση στην πίστα (μέτρα)')
        plt.ylabel('Ταχύτητα (km/h)')
        plt.title('Τηλεμετρία Ταχύτητας - Lewis Hamilton (Silverstone 2024)')
        plt.legend()
        plt.grid(True)

        # Εμφάνιση του γραφήματος
        plt.show()

```

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-----
NotADirectoryError                                Traceback (most
recent call last)
Cell In[2], line 9
      6 plotting.setup_mpl()
      8 # Ενεργοποίηση της προσωρινής μνήμης (cache) για να
κατεβαίνουν τα δεδομένα γρήγορα
----> 9 fastf1.Cache.enable_cache('cache_dir')
      11 # 1. Φορτώνουμε το Grand Prix της Μεγάλης Βρετανίας
2024 (Qualifying)
      12 session = fastf1.get_session(2024, 'Silverstone', 'Q')

File ~\anaconda3\Lib\site-packages\fastf1\req.py:242, in
Cache.enable_cache(cls, cache_dir, ignore_version, force_renew,
use_requests_cache)
      239 cache_dir = os.path.expanduser(cache_dir)
      241 if not os.path.exists(cache_dir):
--> 242     raise NotADirectoryError("Cache directory does not

```

```

exist! Please "
    243                                     "check for typos or create
it first.")
    244 cls._CACHE_DIR = cache_dir
    245 cls._IGNORE_VERSION = ignore_version

NotADirectoryError: Cache directory does not exist! Please
check for typos or create it first.

```

```

In [5]: # 1. Παίρνουμε τον ταχύτερο γύρο του Russell (RUS)
rus_lap = session.laps.pick_driver('RUS').pick_fastest()

# 2. Παίρνουμε τηλεμετρία για τον Russell
rus_telemetry = rus_lap.get_car_data().add_distance()

# 3. Φτιάχνουμε το κοινό γράφημα
plt.figure(figsize=(12, 6))

# Γραμμή για Hamilton (Cyan)
plt.plot(ham_telemetry['Distance'], ham_telemetry['Speed'],
label='Hamilton', color='cyan')
# Γραμμή για Russell (Silver/White)
plt.plot(rus_telemetry['Distance'], rus_telemetry['Speed'],
label='Russell', color='white', linestyle='--')

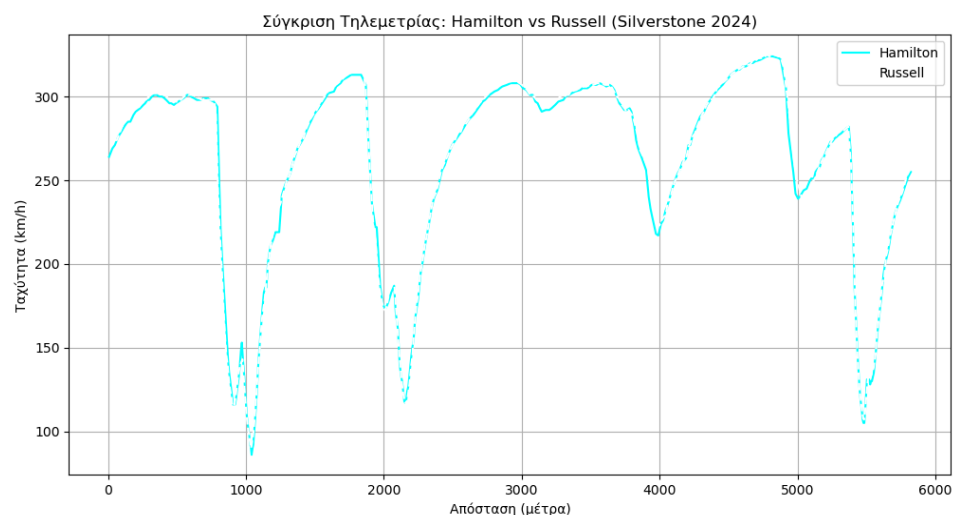
plt.xlabel('Απόσταση (μέτρα)')
plt.ylabel('Ταχύτητα (km/h)')
plt.title('Σύγκριση Τηλεμετρίας: Hamilton vs Russell (Silverstone
2024)')
plt.legend()
plt.grid(True)
plt.show()

```

```

C:\Users\User\anaconda3\Lib\site-packages\fastf1\core.py:3175:
FutureWarning: pick_driver is deprecated and will be removed in
a future release. Use pick_drivers instead.
    warnings.warn("pick_driver is deprecated and will be
removed"

```

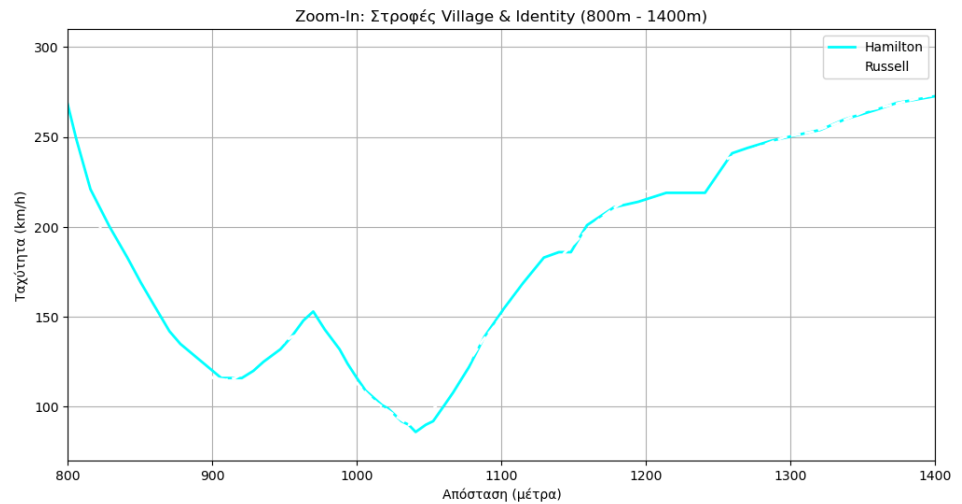



```
In [6]: plt.figure(figsize=(12, 6))

# Σχεδιάζουμε πάλι τις γραμμές
plt.plot(ham_telemetry['Distance'], ham_telemetry['Speed'],
label='Hamilton', color='cyan', linewidth=2)
plt.plot(rus_telemetry['Distance'], rus_telemetry['Speed'],
label='Russell', color='white', linestyle='--', linewidth=2)

# ΕΔΩ ΕΙΝΑΙ ΤΟ ΚΟΛΠΟ: Ορίζουμε τα όρια του άξονα X για να κάνουμε Zoom
στην πρώτη φουρκέτα
plt.xlim(800, 1400)
plt.ylim(70, 310) # Zoom και στον άξονα Y για να φαίνονται οι καθαρές
ταχύτητες στροφής

plt.xlabel('Απόσταση (μέτρα)')
plt.ylabel('Ταχύτητα (km/h)')
plt.title('Zoom-In: Στροφές Village & Identity (800m - 1400m)')
plt.legend()
plt.grid(True)
plt.show()
```

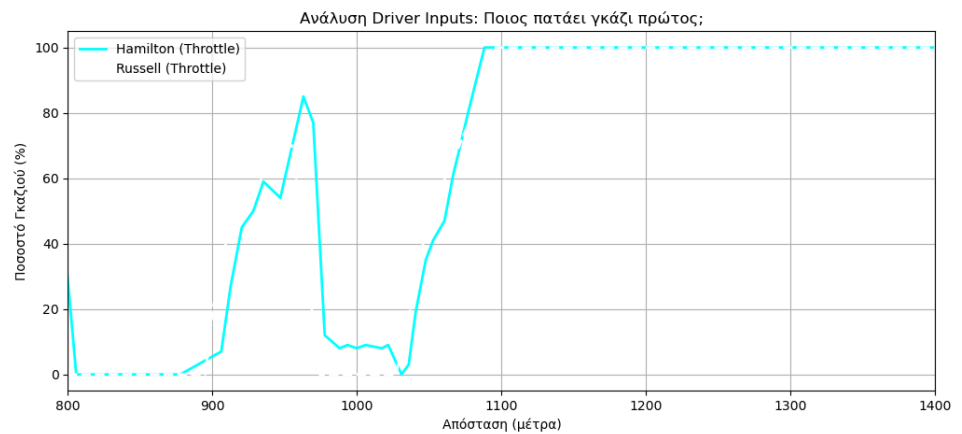


```
In [7]: plt.figure(figsize=(12, 5))

# Σχεδιάζουμε το γκάζι (Throttle) για τους δύο οδηγούς
plt.plot(ham_telemetry['Distance'], ham_telemetry['Throttle'],
label='Hamilton (Throttle)', color='cyan', linewidth=2)
plt.plot(rus_telemetry['Distance'], rus_telemetry['Throttle'],
label='Russell (Throttle)', color='white', linestyle='--', linewidth=2)

# Κάνουμε Zoom ακριβώς στο ίδιο σημείο
plt.xlim(800, 1400)

plt.xlabel('Απόσταση (μέτρα)')
plt.ylabel('Ποσοστό Γκαζιού (%)')
plt.title('Ανάλυση Driver Inputs: Ποιος πατάει γκάζι πρώτος;')
plt.legend()
plt.grid(True)
plt.show()
```



```
In [8]: # Φτιάχνουμε ένα γράφημα με δύο υπο-γραφήματα (ένα για RPM και ένα για Gears)
import matplotlib.pyplot as plt
fig, ax = plt.subplots(2, 1, figsize=(12, 8), sharex=True)

# 1. Πάνω γράφημα: Στροφές Κινητήρα (RPM)
ax[0].plot(ham_telemetry['Distance'], ham_telemetry['RPM'],
label='Hamilton (RPM)', color='cyan')
ax[0].plot(rus_telemetry['Distance'], rus_telemetry['RPM'],
label='Russell (RPM)', color='white', linestyle='--')
ax[0].set_ylabel('Στροφές Κινητήρα (RPM)')
ax[0].title.set_text('Ανάλυση Κινητήρα και Σασμάν (Silverstone 2024)')
ax[0].legend()
ax[0].grid(True)

# 2. Κάτω γράφημα: Σχέση στο Κιβώτιο (Gear)
ax[1].plot(ham_telemetry['Distance'], ham_telemetry['Gear'],
label='Hamilton (Gear)', color='cyan')
ax[1].plot(rus_telemetry['Distance'], rus_telemetry['Gear'],
label='Russell (Gear)', color='white', linestyle='--')
ax[1].set_ylabel('Σχέση Κιβωτίου (1-8)')
ax[1].set_xlabel('Απόσταση (μέτρα)')
ax[1].legend()
ax[1].grid(True)

# Κάνουμε Zoom στο ίδιο ακριβώς σημείο (800m - 1400m)
plt.xlim(800, 1400)

plt.show()
```

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KeyError                                Traceback (most
recent call last)
File ~\anaconda3\Lib\site-
packages\pandas\core\indexes\base.py:3812, in
Index.get_loc(self, key)
    3811 try:
-> 3812     return self._engine.get_loc(casted_key)
    3813 except KeyError as err:

File pandas/_libs/index.pyx:167, in
pandas._libs.index.IndexEngine.get_loc()

File pandas/_libs/index.pyx:196, in
pandas._libs.index.IndexEngine.get_loc()

File pandas/_libs/hashtable_class_helper.pxi:7088, in
pandas._libs.hashtable.PyObjectHashTable.get_item()

File pandas/_libs/hashtable_class_helper.pxi:7096, in
pandas._libs.hashtable.PyObjectHashTable.get_item()

KeyError: 'Gear'
```

The above exception was the direct cause of the following exception:

```

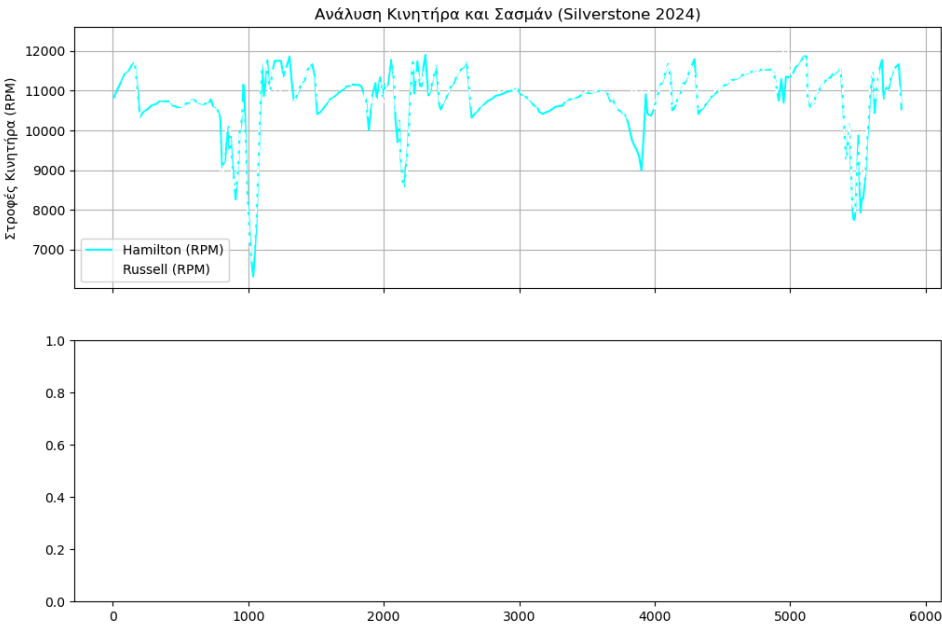
KeyError                                Traceback (most
recent call last)
Cell In[8], line 14
      11 ax[0].grid(True)
      13 # 2. Κάτω γράφημα: Σχέση στο Κιβώτιο (Gear)
--> 14 ax[1].plot(ham_telemetry['Distance'],
ham_telemetry['Gear'], label='Hamilton (Gear)', color='cyan')
      15 ax[1].plot(rus_telemetry['Distance'],
rus_telemetry['Gear'], label='Russell (Gear)', color='white',
linestyle='--')
      16 ax[1].set_ylabel('Σχέση Κιβωτίου (1-8)')

File ~\anaconda3\Lib\site-packages\pandas\core\frame.py:4113,
in DataFrame.__getitem__(self, key)
    4111 if self.columns.nlevels > 1:
    4112     return self._getitem_multilevel(key)
-> 4113 indexer = self.columns.get_loc(key)
    4114 if is_integer(indexer):
    4115     indexer = [indexer]

File ~\anaconda3\Lib\site-
packages\pandas\core\indexes\base.py:3819, in
Index.get_loc(self, key)
    3814     if isinstance(casted_key, slice) or (
    3815         isinstance(casted_key, abc.Iterable)
    3816         and any(isinstance(x, slice) for x in
casted_key)
    3817     ):
    3818         raise InvalidIndexError(key)
-> 3819     raise KeyError(key) from err
    3820 except TypeError:
    3821     # If we have a listlike key, _check_indexing_error
will raise
    3822     # InvalidIndexError. Otherwise we fall through and
re-raise
    3823     # the TypeError.
    3824     self._check_indexing_error(key)

KeyError: 'Gear'

```



```
In [9]: import matplotlib.pyplot as plt

# 1. Παίρνουμε την ΠΛΗΡΗ τηλεμετρία (get_telemetry) που περιέχει και τα
Gears
ham_telemetry_correct = ham_lap.get_telemetry().add_distance()
rus_telemetry_correct = rus_lap.get_telemetry().add_distance()

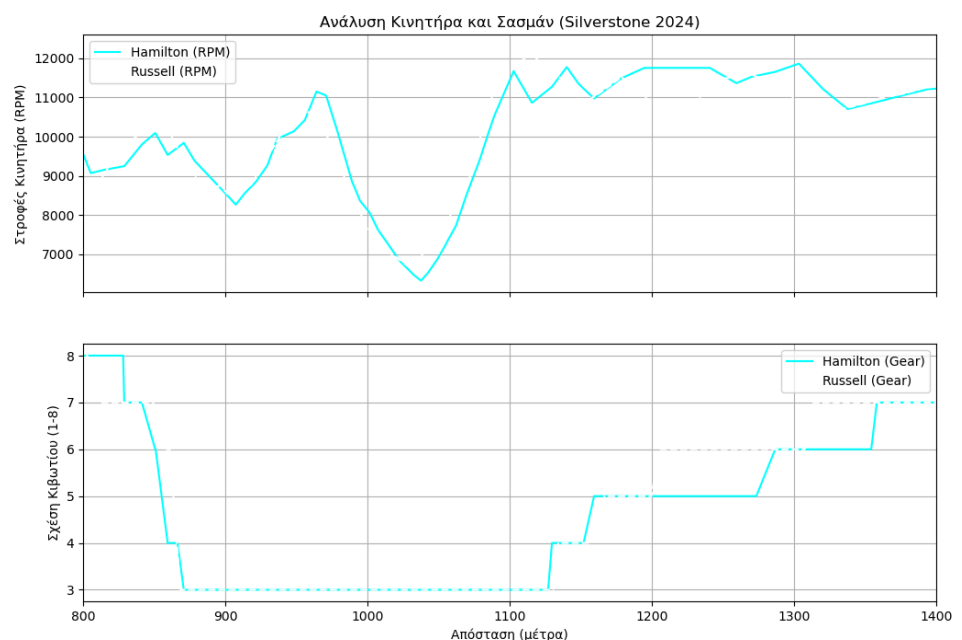
# 2. Φτιάχνουμε τα γραφήματα
fig, ax = plt.subplots(2, 1, figsize=(12, 8), sharex=True)

# Πάνω γράφημα: Στροφές Κινητήρα (RPM)
ax[0].plot(ham_telemetry_correct['Distance'],
ham_telemetry_correct['RPM'], label='Hamilton (RPM)', color='cyan')
ax[0].plot(rus_telemetry_correct['Distance'],
rus_telemetry_correct['RPM'], label='Russell (RPM)', color='white',
linestyle='--')
ax[0].set_ylabel('Στροφές Κινητήρα (RPM)')
ax[0].title.set_text('Ανάλυση Κινητήρα και Σασμάν (Silverstone 2024)')
ax[0].legend()
ax[0].grid(True)

# Κάτω γράφημα: Σχέση στο Κιβώτιο (nGear) - Στο πλήρες telemetry
ονομάζεται nGear!
ax[1].plot(ham_telemetry_correct['Distance'],
ham_telemetry_correct['nGear'], label='Hamilton (Gear)', color='cyan')
ax[1].plot(rus_telemetry_correct['Distance'],
rus_telemetry_correct['nGear'], label='Russell (Gear)', color='white',
linestyle='--')
ax[1].set_ylabel('Σχέση Κιβωτίου (1-8)')
ax[1].set_xlabel('Απόσταση (μέτρα)')
ax[1].legend()
ax[1].grid(True)

# Κάνουμε Zoom στο ίδιο ακριβώς σημείο (800m - 1400m)
plt.xlim(800, 1400)

plt.show()
```



In []:

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